

Set Theory: Fundamentals and Applications

Sets, Operations, Inclusion–Exclusion & Hasse Diagrams

Ahmed Sultan P

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Outline

- 1 Sets
- 2 Set Builder Form
- 3 Types of Sets
- 4 Set Operations
- 5 Inclusion–Exclusion Principle
- 6 Hasse Diagrams

What is a Set?

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Examples

- $V = \{a, e, i, o, u\}$ — vowels of the English alphabet
- $\mathbb{N} = \{1, 2, 3, \dots\}$ — natural numbers
- $E = \{2, 4, 6, 8, \dots\}$ — even natural numbers

Representation of Sets

Two common ways to represent a set:

- 1 **Roster (Tabular) Form:** List all elements, separated by commas, inside braces.

$$A = \{1, 2, 3, 4, 5\}$$

- 2 **Set Builder Form:** Describe elements by a common property (next section).

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Note

- Order of elements does not matter: $\{1, 2, 3\} = \{3, 1, 2\}$
- Repetition is ignored: $\{1, 1, 2\} = \{1, 2\}$

Set Builder Form

Definition

In **set builder form**, a set is described by stating a property $P(x)$ that its elements satisfy:

$$A = \{ x \mid P(x) \} \quad \text{or} \quad A = \{ x : P(x) \}$$

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Examples

- $A = \{ x \mid x \in \mathbb{N}, x \leq 5 \} = \{1, 2, 3, 4, 5\}$
- $B = \{ x \mid x \text{ is a vowel of the English alphabet} \} = \{a, e, i, o, u\}$
- $C = \{ x \in \mathbb{Z} \mid x^2 = 4 \} = \{-2, 2\}$
- $D = \{ 2n \mid n \in \mathbb{N} \} = \{2, 4, 6, \dots\}$

Roster Form vs. Set Builder Form

Roster Form	Set Builder Form
$\{1, 4, 9, 16, 25\}$	$\{x^2 \mid x \in \mathbb{N}, x \leq 5\}$
$\{2, 3, 5, 7, 11, 13\}$	$\{x \mid x \text{ is a prime, } x < 15\}$
$\{\dots, -2, -1, 0, 1, 2, \dots\}$	$\{x \mid x \in \mathbb{Z}\}$

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- Set builder form is essential for *infinite* sets or sets whose elements are inconvenient to list.
- Roster form is convenient for small, finite sets.

Types of Sets (I)

Empty (Null) Set A set with no elements; denoted $\emptyset$ or $\{\}$.

Example: $\{x \in \mathbb{N} \mid x < 0\} = \emptyset$

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Infinite Set A set that is not finite.

Example: $\mathbb{N}, \mathbb{Z}, \mathbb{R}$

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Power Set $\mathcal{P}(A)$: the set of all subsets of A .

If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

Example: $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

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Universal Set The set U containing all objects under discussion.

Disjoint Sets A and B are disjoint iff $A \cap B = \emptyset$.

Set Operations

Let A and B be subsets of a universal set U .

Union $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Difference $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Complement $A' = \bar{A} = \{x \in U \mid x \notin A\}$

Symmetric Difference $A \triangle B = (A - B) \cup (B - A)$

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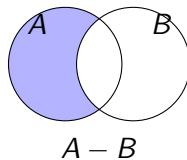
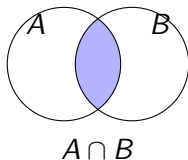
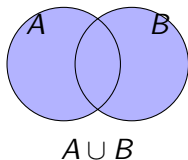
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Example: $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $U = \{1, \dots, 8\}$

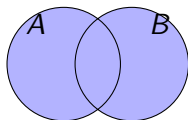
$A \cup B = \{1, 2, 3, 4, 5, 6\}$, $A \cap B = \{3, 4\}$,

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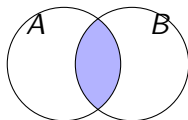
Venn Diagrams of Basic Operations



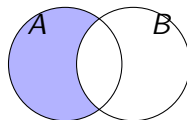
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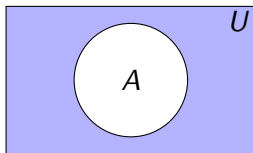
$$A \cup B$$



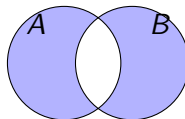
$$A \cap B$$



$$A - B$$



$$A'$$



$$A \triangle B$$

Laws of Set Algebra

Law	Statement
Idempotent	$A \cup A = A, \quad A \cap A = A$
Commutative	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A, \quad A \cap U = A$
Complement	$A \cup A' = U, \quad A \cap A' = \emptyset$
De Morgan's	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

Inclusion–Exclusion Principle

For two sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

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For three sets

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |B \cap C| - |A \cap C| \\ &\quad + |A \cap B \cap C| \end{aligned}$$

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Idea: Adding $|A|$ and $|B|$ counts elements of $A \cap B$ *twice*, so we subtract the overlap once.

General Form and Example

General form (n sets)

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|$$

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Example

In a class of 50 students, 30 study Mathematics, 25 study Physics, and 10 study both. How many study at least one subject?

$$|M \cup P| = |M| + |P| - |M \cap P| = 30 + 25 - 10 = 45$$

Students studying neither: $50 - 45 = 5$.

Partial Orders and Hasse Diagrams

Partially Ordered Set (Poset)

A relation $\preceq$ on a set S is a **partial order** if it is

- **Reflexive:** $a \preceq a$
- **Antisymmetric:** $a \preceq b$ and $b \preceq a \Rightarrow a = b$
- **Transitive:** $a \preceq b$ and $b \preceq c \Rightarrow a \preceq c$

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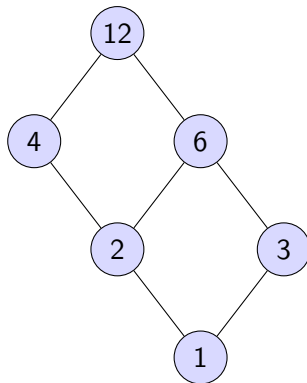
Hasse Diagram

A simplified graph of a finite poset, obtained by:

- 1 Removing all self-loops (reflexivity is implicit).
- 2 Removing edges implied by transitivity.
- 3 Drawing b above a whenever $a \prec b$ (arrows omitted; direction is upward).

Example 1: Divisibility on $\{1, 2, 3, 4, 6, 12\}$

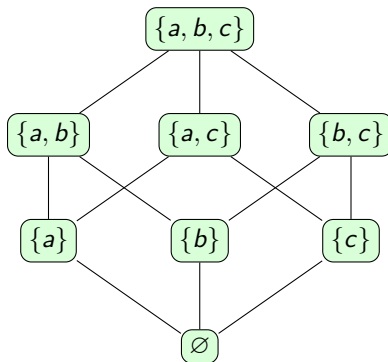
Poset: $(\{1, 2, 3, 4, 6, 12\}, |)$ where $a | b$ means “ a divides b ”.



Edge 1–4 is omitted because $1 | 2$ and $2 | 4$ (transitivity).

Example 2: Power Set of $\{a, b, c\}$ under $\subseteq$

Poset: $(\mathcal{P}(\{a, b, c\}), \subseteq)$



Reading a Hasse Diagram

- **Maximal element:** no element lies above it.
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Remember

Comparability in a Hasse diagram means there is an *upward path* between two elements — not merely that one is drawn higher.

Summary

- A **set** is a well-defined collection of distinct objects; written in roster or **set builder** form.
- Key **types**: empty, singleton, finite/infinite, equal/equivalent, subsets, power set, universal, disjoint.
- **Operations**: union, intersection, difference, complement, symmetric difference — governed by the laws of set algebra (incl. De Morgan's laws).
- **Inclusion–Exclusion** counts a union without double-counting overlaps.
- **Hasse diagrams** give a compact picture of finite posets by dropping reflexive and transitive edges.

Thank you!